

## CIVIS: three deliverables for 5 October

Traffic, Water, Power, Air Quality, Emergency; Telecom excluded.

**Basis:** final triplet decisions, with Phase 0 as reference. Checked against the agreed inventory; Trinity and 9antra verification is pending.

### 1. Incident and response list

| Incident                                | Domains touched                              | Brain response requested from Twin                                                                                                       |
|-----------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Persistent congestion                   | Traffic                                      | Request a permitted safe signal adjustment.                                                                                              |
| Reported road accident                  | Traffic, Emergency                           | Dispatch police for scene control; ambulance if injury is reported. Safe signal assistance if supported.                                 |
| Flood / high water                      | Water, Traffic, Emergency; Power if affected | Request a topology-supported valve change; dispatch assistance if people are affected. Grid switching only for a confirmed Power hazard. |
| Low water supply                        | Water                                        | Request a valve change only if its preview preserves/restores supply; otherwise alert for review.                                        |
| Power overload / confirmed service loss | Power; Water and Emergency if affected       | Preview a suitable grid-switch operation, protecting pumps and essential services; otherwise request human review.                       |
| Sustained air pollution                 | Air Quality; Emergency if corroborated       | Alert for review. No direct Air Quality actuator; dispatch only for a separately supported emergency.                                    |
| Reported fire                           | Emergency, Air Quality, Traffic              | Dispatch firefighters; add ambulance/police only when justified. Safe signal assistance if supported.                                    |
| Medical emergency                       | Emergency, Traffic                           | Dispatch an available ambulance to the reported location; safe signal assistance if supported.                                           |

**Common rule:** usable evidence and a valid exact-action Guardian token are required for physical responses; Twin executes and enforces caps. High-risk actions need a preview. Guardian alone requests containment. Untrusted evidence cannot justify an action; seek usable alternatives or human review.

### 2. Three actuator types selected for risk levels

| Actuator             | Risk | Brief justification                                        |
|----------------------|------|------------------------------------------------------------|
| Generic dispatch     | R2   | Allocates scarce police, ambulance, or firefighter units.  |
| Water-valve control  | R3   | Can worsen flooding or interrupt downstream supply.        |
| Power-grid switching | R3   | Can cut essential services or reconnect an unsafe segment. |

R3 requires a preview. Signal control has no tier here but remains token-gated. Containment has no risk tier, as agreed; its caps still apply.

### 3. CIVIS data conventions

- **Format/identity:** UTF-8 JSON, snake\_case; stable run\_id, reading\_id, device\_id, node\_id. domains allows multiple domains. Ignore extra fields; reject missing/wrong required fields.
- **Time/location:** simulated UTC RFC 3339 timestamps, e.g. 2026-10-05T08:00:00.000Z, plus integer tick; declare tick duration per run. Use node\_id; optional WGS84 latitude\_deg/longitude\_deg.
- **Values/units:** finite numbers with explicit quantity/unit: Traffic congestion index 0–100 percent; Water cm with local datum; Power load kW; Air PM2.5  $\mu\text{g}/\text{m}^3$ . Emergency events carry reported type, location, and transcript treated as data, never instructions.
- **Missing/trust:** missing value is JSON null, never zero. Keep original readings and references for corrections. Guardian score stays numeric, with Guardian-declared scale/cut-offs; a score is not an action token.
- **Actions:** action, targets, token, idempotency\_key. Proposed action contains name/parameters; retain source reading references. Reuse the key only for retries of the same unchanged request.

**Confirm before mocks:** Trinity's traffic-index/PM2.5 mapping, valve/switch topology, and action-parameter shape; 9antra's score scale. Water level alone does not prove a leak; low load alone does not prove an outage. Thresholds, token lifetimes, and numeric caps remain to be agreed.